

Intro to electronics

Task 1

1. Identifying the needs

- HC-SR04 Ultrasonic Sensor: 5V, 15mA operating current.
- Re22 relay: needs 9V, the 9V coil draws 50mA (0.45W standard type, 180Ω coil).
- 3S Li-ion battery (12.6V)

NOTE: 3S Li-ion is 12.6V when full but goes down towards 9 to 11V because it drains. Any regulator needs its input to stay comfortably above its output. It will be hard just to add a 9V battery as it can also go down and our relay needs 9V.

2. Choose the components:

The core call is LDO vs buck for each rail:

- LDO: linear, simple, quiet, cheap. It burns the extra voltage as heat, equal to $(V_{in} - V_{out}) \times I$. Good when current is low or the voltage drop is small.
- Buck: switching, efficient, stays cool even with a big drop or high current. More complex and adds some electrical noise.

For sensor: (5V, 15mA): big drop but tiny current. Heat is only $7.6V \times 0.015A = 0.11W$, negligible. A clean output is nice for a sensor too. LDO is the sensible pick.

For Relay: only a 3.6V drop. If the coil pulls tens of mA, an LDO is fine. If it pulls hundreds of mA, the LDO runs hot and a buck is the better choice. The coil pulls 50mA, so an LDO is fine.

Protection circuit:

1. For the relay: put a flyback diode across the coil. It is an inductive load, and switching it off creates a voltage spike that will damage parts without one.

2. Option 1: Series Schottky diode:

The downside is a diode always drops some voltage when conducting. A Schottky drops less than a normal diode, about 0.3 to 0.4V, but it is still a loss

3. Option 2: P-channel MOSFET

The big advantage is that when it is on, it behaves like a tiny resistor, not a diode. The voltage lost is nearly zero, often just a few millivolts. So you keep almost all your battery voltage and waste almost no power.

The downside is it is a few more parts (the MOSFET plus a resistor and usually a small Zener to protect the gate) and slightly more.

- P-CHANNEL MOSFET would be more suitable, but both options are fine.

3. Check the datasheet

- For the HC-SR04:

Working voltage: 5V

Working current: 15mA

Step 3 asks to check the datasheet of the regulator I picked

- 5V rail (HC-SR04 sensor, 5V / 15mA) — LM7805

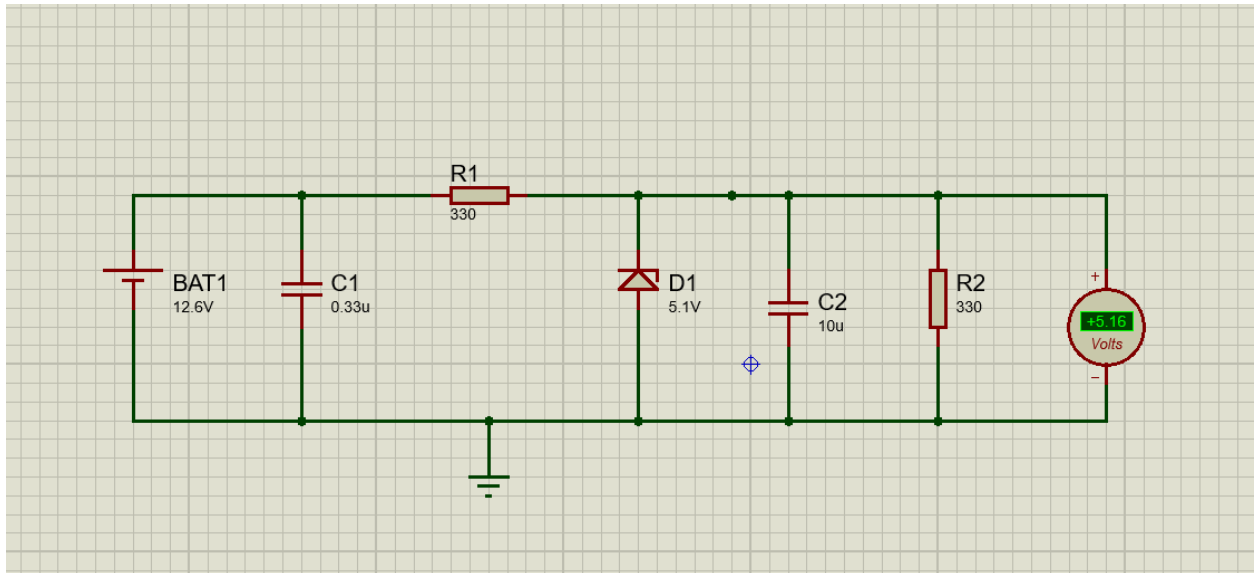
The LM7805 is a fixed 5V linear regulator. Its recommended input range is 7V to 35V, so the 12.6V battery sits comfortably inside it, and its rated output current is up to 1.5A, far above the 15mA the sensor draws. Its dropout voltage is around 2V, meaning the input must stay above roughly 7V to hold 5V; even with battery sag this rail is never at risk.

- V rail (Songle SRD relay, 9V / 50mA) — LM2940-9.0

A standard 7809 was rejected because it needs about 2V of headroom, so it would lose regulation as the 3S battery sags. The LM2940-9.0 is a 9V low-dropout regulator instead. It sources up to 1A with a dropout voltage of typically 0.5V (1V max over temperature), well above the 50mA coil current. It regulates 9V for input from 10.5V up to 26V, so the full 12.6V battery is covered, and the low dropout keeps the 9V rail alive further down the discharge curve than a 7809 could. The LM2940-9.0 therefore fits: input range covers the battery, current rating exceeds the load, and its low dropout suits the sagging battery.

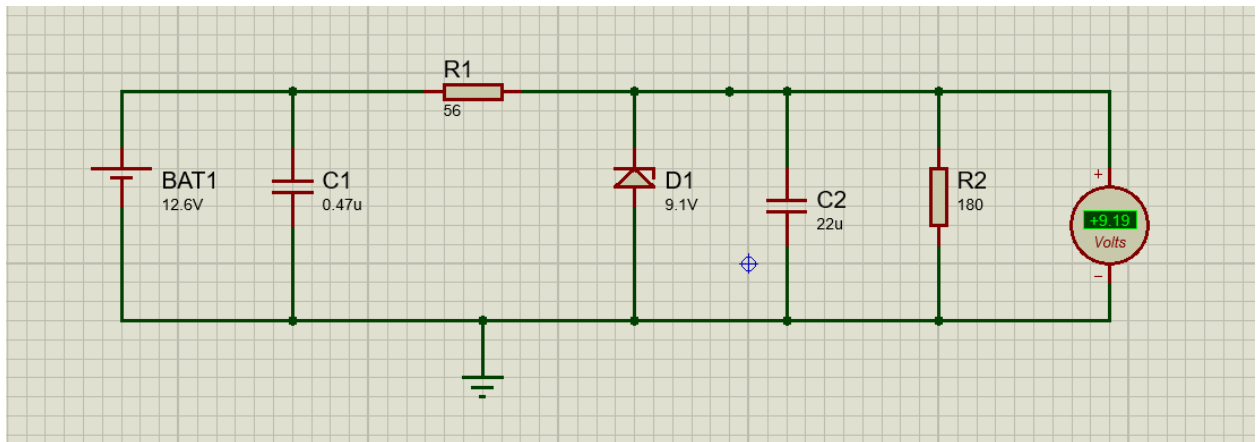
4. Simulation on proteus:

The 5V Circuit (LM7805)



The circuit takes the 12.6V battery and produces a steady 5V for the HC-SR04 sensor. A series resistor R_s (330Ω) carries the current from the battery to the output node. A 5.1V Zener diode (D1) is connected across that node to ground. When the node tries to rise above 5.1V, the Zener conducts in reverse breakdown and holds the voltage fixed at its rated value, diverting any surplus current to ground. The input and output capacitors ($0.33\mu\text{F}$ and $10\mu\text{F}$) smooth the supply and keep the output stable. The 330Ω load resistor (R2) represents the sensor, since 5V divided by its 15mA draw equals about 330Ω . The simulation confirmed the output settled and held at roughly 5V under this load.

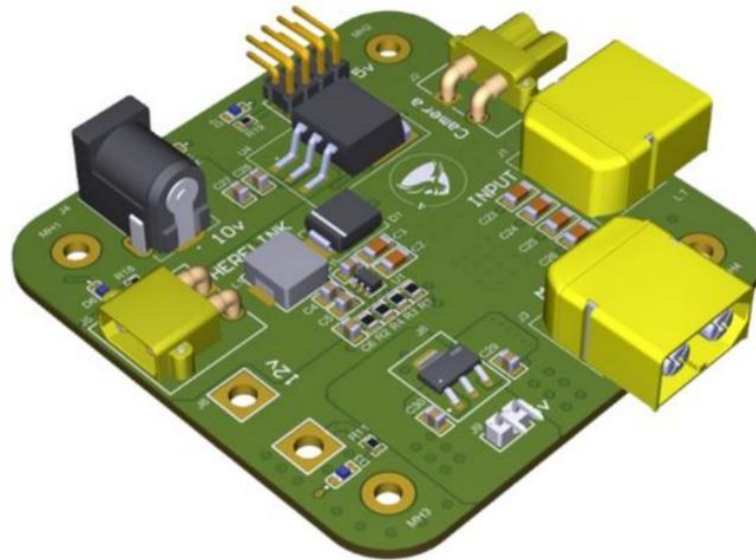
The 9V circuit (LM2940-9.0)



This circuit works the same way but is tuned for the 9V relay. R_s is reduced to 56Ω because the relay draws more current than the sensor, so a smaller resistor is needed to feed the higher demand. A 9.1V Zener sets the output voltage. The output capacitor is increased to $22\mu\text{F}$ for smoothing, and the 180Ω load resistor represents the relay coil, since 9V divided by its 50mA draw equals 180Ω . The simulation confirmed a stable output at roughly 9V.

Why a Zener, not the regulator ICs: The chosen design uses linear regulators (LM7805 and LM2940). For simulation, a simpler Zener shunt was used because it builds from standard library parts and clearly shows the key behaviour: a stable regulated output from a higher input. It is less efficient than a regulator IC, so it serves

Task 2



This is a small multi-rail power/distribution board (silkscreen shows 5V, 10V, 12V rails, an "INPUT", and a "Camera" connector), so most parts are there to condition and regulate power.

Capacitor (C2) stores small bits of charge to smooth the voltage and soak up noise. Needed so the power rails stay steady when chips suddenly pull current.

Resistor (R11) limits current and sets voltage levels. Needed to protect parts from too much current and to tell the regulators what voltage to output.

Inductor (L1) stores energy in a magnetic field. The big shielded one does the heavy lifting in the switching regulator (stepping voltage up or down efficiently); the small blue one just filters noise.

Voltage Regulator (U4) takes a messy input voltage and gives a clean fixed output. Needed because circuits only work properly on steady voltages, and the input can wobble.

Connector (J4) is the DC power jack, your way of getting power onto the board. The yellow screw terminals and camera connector are the other connectors. Needed so you can plug things in without soldering.

Diode (D1) only lets current flow one way. Here it's likely for reverse-polarity protection (saves the board if you plug power in backwards) and to give the inductor a path to dump its energy in the switching circuit.